KINGDOM OF CORPUS · STUDY NOTES

Land 7: Synapse Caverns

Nervous tissue and the CNS · OpenStax Ch. 12 and 13

Key ideas

Organization of the nervous system

- CNS is the brain and spinal cord; PNS is cranial nerves, spinal nerves and ganglia.
- Somatic system: conscious sensation and voluntary control of skeletal muscle.
- Autonomic system: involuntary control of cardiac muscle, smooth muscle and glands.
- Enteric system: neurons in the wall of the gut that control motility and secretion.
- Gray matter holds cell bodies and synapses; white matter holds myelinated axon tracts.

Neurons and glia

- Dendrites receive input, the axon hillock (initial segment) starts action potentials, axon terminals release transmitter.
- Multipolar (one axon, many dendrites) is most common; bipolar in retina and olfactory epithelium; unipolar sensory neurons have cell bodies in ganglia.
- Sensory (afferent) neurons carry signals in, motor (efferent) neurons carry them out, interneurons connect them inside the CNS.
- CNS glia: astrocytes (blood brain barrier, ion balance), oligodendrocytes (CNS myelin), microglia (phagocytes), ependymal cells (line ventricles, help make CSF).
- PNS glia: Schwann cells (PNS myelin) and satellite cells (support cell bodies in ganglia).
- Multiple sclerosis destroys CNS myelin; Guillain Barre syndrome attacks PNS myelin.

Resting and graded potentials

- Resting membrane potential is about negative 70 mV, set mainly by K+ leaking out.
- The Na+/K+ pump moves 3 Na+ out and 2 K+ in per ATP to keep gradients.
- Graded potentials vary with stimulus size, fade with distance and can summate.
- EPSP: depolarizing, usually Na+ entry. IPSP: hyperpolarizing, Cl entry or K+ exit.
- Temporal summation: one synapse firing rapidly. Spatial summation: many synapses at once.

Action potentials

- Threshold is about negative 55 mV; the peak is about positive 30 mV.
- Depolarization: voltage gated Na+ channels open and Na+ rushes in.
- Repolarization: voltage gated K+ channels open and K+ rushes out; slow K+ closing causes the brief hyperpolarization.
- All or none: every action potential is full size; stronger stimuli raise firing frequency, not amplitude.
- Absolute refractory period (Na+ channels inactivated): no new spike; relative refractory period: only a stronger stimulus works. This keeps conduction one way.
- Speed rises with axon diameter and myelin; saltatory conduction jumps between nodes of Ranvier.

Synapses and neurotransmitters

- Chemical synapse: action potential opens voltage gated Ca2+ channels, vesicles release transmitter, which binds receptors after a short delay.
- Electrical synapses use gap junctions for direct, fast, often two way current.
- Ionotropic receptors are ion channels (fast); metabotropic receptors use G proteins (slower).
- Glutamate is the main excitatory CNS transmitter; GABA (brain) and glycine (spinal cord) are the main inhibitory ones.
- Acetylcholine is cleared by acetylcholinesterase; norepinephrine, dopamine and serotonin are biogenic amines cleared mostly by reuptake.
- Loss of dopamine neurons in the substantia nigra causes Parkinson disease; SSRIs block serotonin reuptake.

Brain development and the cerebrum

- Primary vesicles: prosencephalon, mesencephalon, rhombencephalon.
- Secondary vesicles: telencephalon (cerebrum), diencephalon (thalamus, hypothalamus), mesencephalon (midbrain), metencephalon (pons, cerebellum), myelencephalon (medulla).
- Frontal lobe: precentral gyrus (primary motor cortex), Broca's area, planning. Parietal lobe: postcentral gyrus (somatosensory).
- Temporal lobe: hearing, Wernicke's area for language comprehension, memory. Occipital lobe: vision.
- Each hemisphere controls and senses the opposite side of the body; the corpus callosum links the hemispheres.
- Basal nuclei start wanted movements and suppress unwanted ones.

Diencephalon, brainstem and cerebellum

- Thalamus: relay to cortex for all senses except smell.
- Hypothalamus: temperature, hunger, thirst, autonomic control and the pituitary.
- Midbrain: superior colliculi (visual reflexes), inferior colliculi (auditory reflexes), substantia nigra, nuclei of III and IV.
- Pons: relays to cerebellum, helps regulate breathing, nuclei of V to VIII.
- Medulla: cardiovascular and respiratory centers, pyramids, nuclei of IX to XII.
- Cerebellum: compares commands with feedback; damage causes same side ataxia and intention tremor.

Protection: meninges, CSF, blood brain barrier

- Meninges outside in: dura, arachnoid, pia (DAP). Dural sinuses lie between the two dura layers.
- CSF is made by choroid plexuses, fills the subarachnoid space and is reabsorbed by arachnoid granulations.
- Flow: lateral ventricles, interventricular foramina, third ventricle, cerebral aqueduct, fourth ventricle, subarachnoid space and central canal.
- Blocking the cerebral aqueduct enlarges the lateral and third ventricles (hydrocephalus).
- Blood brain barrier: tight junctions of capillary endothelium supported by astrocytes; lipid soluble molecules cross easily.

Spinal cord and peripheral nerves

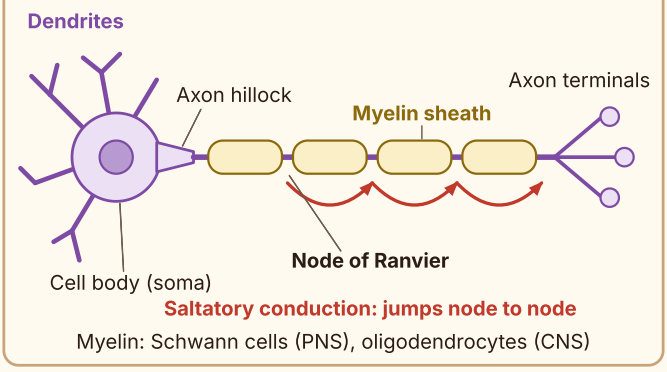
- Adult cord ends near L1 to L2; lumbar puncture below this samples CSF from the subarachnoid space.
- Dorsal horn: sensory input. Ventral horn: somatic motor neurons. Lateral horn (T1 to L2): sympathetic preganglionic neurons.
- Dorsal root is sensory with a dorsal root ganglion; ventral root is motor; spinal nerves are mixed.
- White matter columns carry ascending and descending tracts.
- 31 pairs of spinal nerves; ventral rami form plexuses: cervical (about C1 to C4, phrenic nerve), brachial (C5 to T1), lumbar, sacral (sciatic nerve).
- 12 cranial nerves; key tests: VII facial expression, VIII hearing and balance, X parasympathetic to viscera, XII tongue (deviates toward the damaged side).

Villager wisdom

- Down here in the caverns we say: sodium rushes in to rise, potassium rushes out to fall. That is the action potential in one breath.
- Myelin is like the lantern posts along our tunnels. The signal leaps from node to node, which is why myelinated axons carry messages so fast.
- Remember the meninges from outside in as DAP: dura, arachnoid, pia. Cerebrospinal fluid flows in the space just beneath the arachnoid.
- The left side of the brain steers the right side of the body. Commands to move start in the precentral gyrus, and touch is felt in the postcentral gyrus.
- Graded potentials are whispers that fade with distance, but enough whispers added together at the axon hillock can reach threshold and set off an action potential.

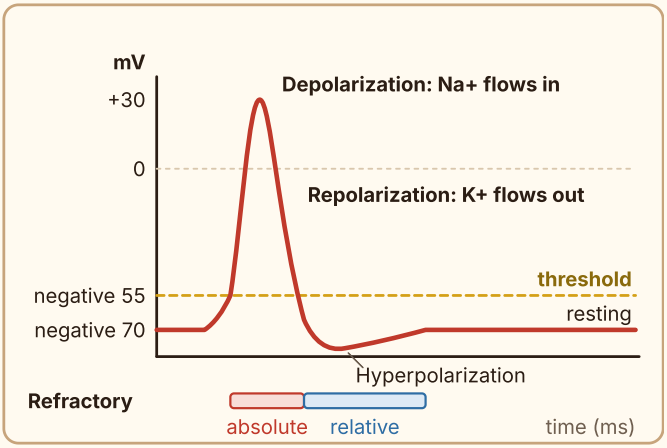
Diagrams

Neuron structure



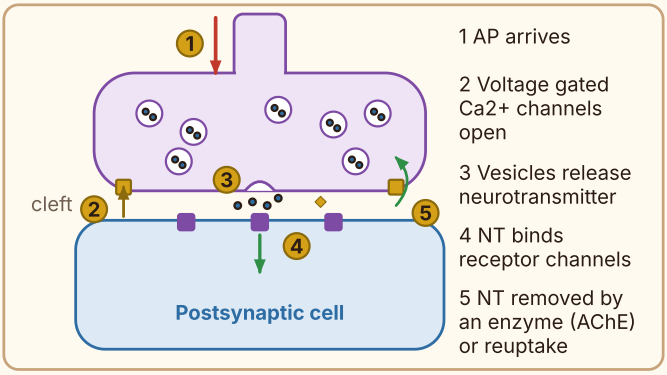
Signals enter at dendrites, start at the axon hillock, and on a myelinated axon jump from node to node (saltatory conduction), which is much faster.

The action potential



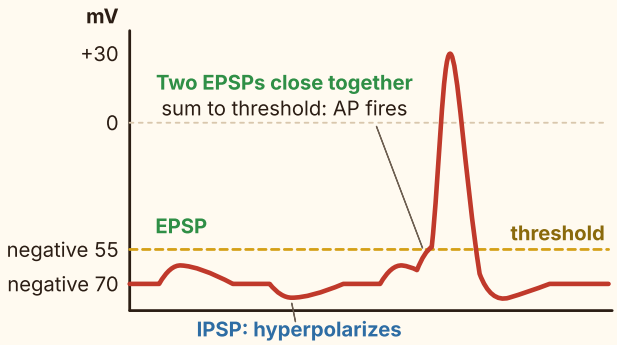
At threshold, voltage gated Na+ channels open (depolarization to about +30 mV); then K+ flows out (repolarization) and briefly overshoots (hyperpolarization).

Chemical synapse



An action potential opens voltage gated Ca2+ channels, Ca2+ triggers exocytosis of neurotransmitter, it binds receptor channels, and enzymes or reuptake clear it from the cleft.

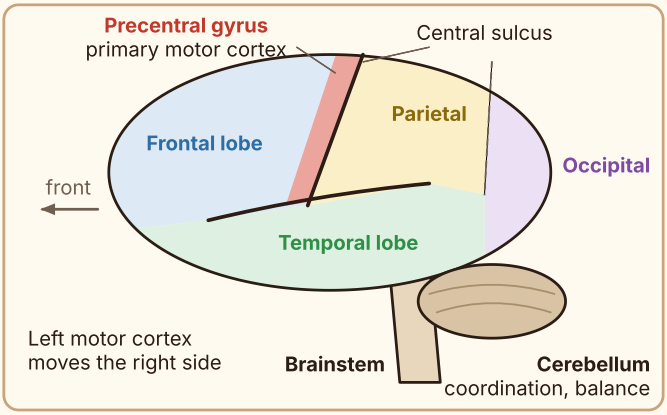
Graded potentials and summation



Example: two 8 mV EPSPs add to 16 mV, past the 15 mV gap

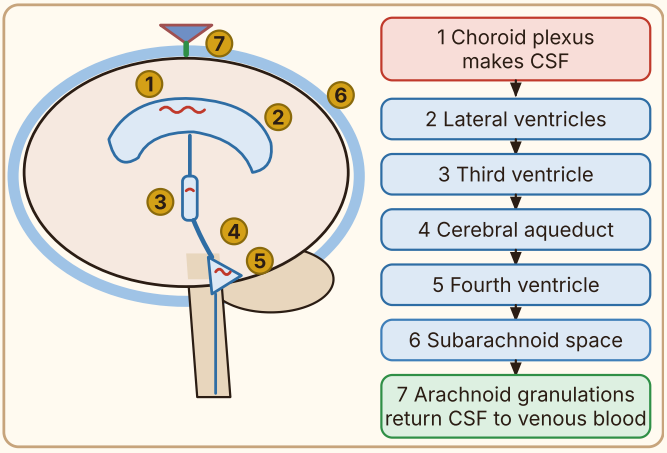
Graded potentials vary in size and fade with distance; EPSPs depolarize and IPSPs hyperpolarize, and only when summed EPSPs reach threshold at the axon hillock does an action potential fire.

Regions of the brain



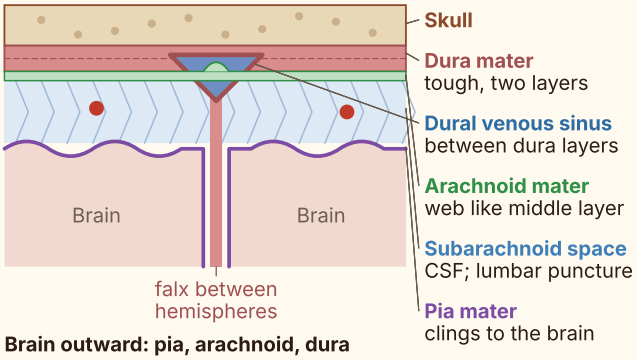
The precentral gyrus (primary motor cortex) controls the opposite side of the body, and the cerebellum coordinates smooth, accurate movement and balance.

Flow of cerebrospinal fluid



Choroid plexuses make CSF in the ventricles; it flows from the lateral to the third ventricle, through the cerebral aqueduct to the fourth, out into the subarachnoid space, and back to venous blood through arachnoid granulations.

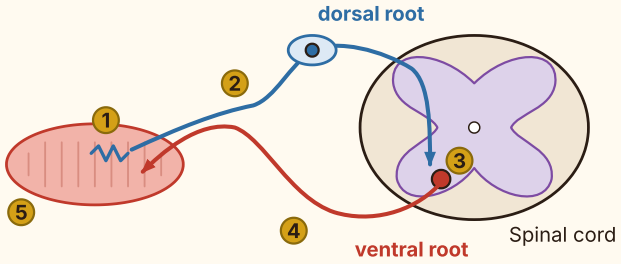
The meninges



Brain outward: pia, arachnoid, dura

From the brain outward the meninges are pia, arachnoid and dura mater; CSF fills the subarachnoid space, where a lumbar puncture samples it, and venous sinuses lie between the two dura layers.

Reflex arc and spinal cord



1 Receptor (stretch)
2 Sensory neuron
3 One synapse in the gray matter
4 Motor neuron
5 Effector muscle

A reflex arc runs receptor, sensory neuron, integration center, motor neuron, effector; sensory fibers enter through the dorsal root and motor fibers leave through the ventral root.

The twelve cranial nerves

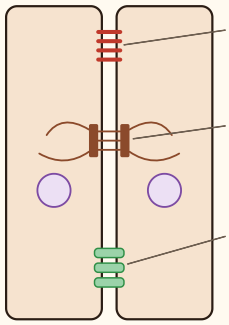
Sensory (blue), motor (red), both (purple)

I Olfactory smell	**VII Facial** expression, eyelid, taste
II Optic vision	**VIII Vestibulocochlear** hearing and balance
III Oculomotor most eye moves, lid, pupil	**IX Glossopharyngeal** taste, swallowing, gag
IV Trochlear superior oblique muscle	**X Vagus** heart, gut, swallow, voice
V Trigeminal face sensation, chewing	**XI Accessory** neck: SCM and trapezius
VI Abducens lateral rectus: eye outward	**XII Hypoglossal** tongue movement

XII lesion: the stuck out tongue deviates toward the injured side

Cranial nerves I to XII are sensory, motor or mixed; for example the facial nerve (VII) closes the eyelid and makes facial expressions, and the abducens (VI) turns the eye outward.

Cell junctions and the blood brain barrier

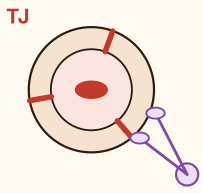


Tight junction
watertight seal, blocks leaks

Desmosome
anchors cells, resists pulling

Gap junction
connexon pores: ions flow, cells act as a unit

Blood brain barrier



Tight junctions seal brain capillaries; only lipid soluble drugs pass

Tight junctions seal neighboring cells, desmosomes rivet them against pulling, and gap junctions (connexons) let ions pass so cells act together; tight junctions in brain capillaries form the blood brain barrier.

Major neurotransmitters

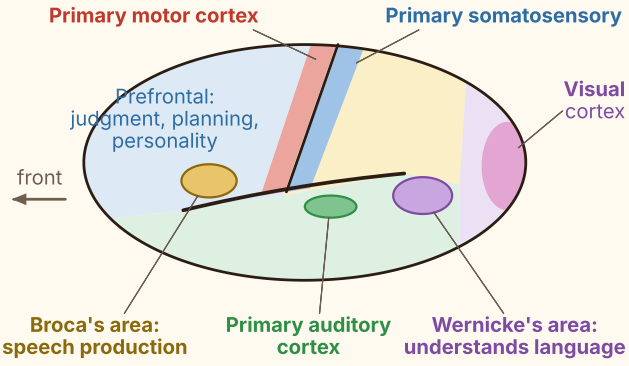
Transmitter	Effect	Key role
Acetylcholine	varies	NMJ, parasympathetic
Glutamate	excites	main excitatory in CNS
GABA	inhibits	main inhibitory in brain
Glycine	inhibits	inhibitory in spinal cord
Dopamine	varies	movement; lost in Parkinson
Norepinephrine	varies	sympathetic, alertness
Serotonin	varies	mood, sleep, appetite

Ionotropic receptor
is itself an ion channel: fast

Metabotropic receptor
G protein and second messenger: slower

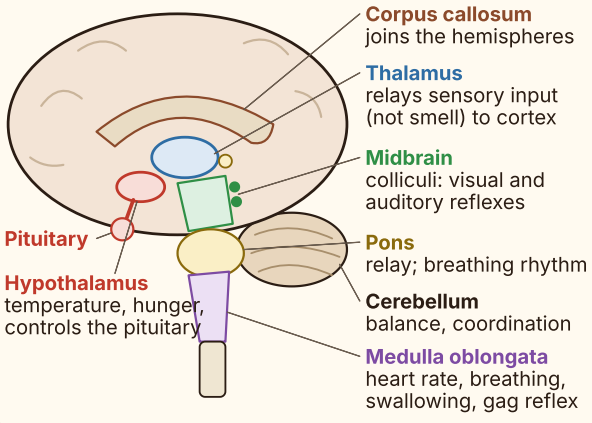
Glutamate is the main excitatory and GABA the main inhibitory transmitter in the brain, glycine inhibits in the spinal cord, and dopamine loss in the substantia nigra causes Parkinson disease.

Functional areas of the cerebral cortex



Broca's area in the frontal lobe plans the movements of speech, while Wernicke's area near the temporal lobe lets us understand language; the precentral gyrus moves the body and the postcentral gyrus feels it.

Diencephalon and brainstem



The diencephalon is the thalamus (sensory relay to the cortex) and hypothalamus (temperature, hunger, pituitary control); the brainstem is the midbrain, pons and medulla, which runs heart rate, breathing and swallowing reflexes.

Self check

1. **The basal nuclei mainly function to:**
    A. Relay sensory input to the cortex
    B. Control heart rate and breathing
    C. Initiate and regulate movement
    D. Produce cerebrospinal fluid
2. **Which axon would conduct an action potential the fastest?**
    A. A thin unmyelinated pain fiber
    B. A thick myelinated motor axon
    C. A thin, lightly myelinated autonomic fiber
    D. A thick axon whose myelin was destroyed
3. **Which factor contributes MOST to the negative resting membrane potential of a neuron?**
    A. K+ leaking out through leak channels
    B. Na+ leaking in through leak channels
    C. Ca2+ pumped into the cell
    D. Chloride ions pumped out of the cell
4. **Broca's area, which plans the motor movements of speech, is located in the:**
    A. Temporal lobe
    B. Occipital lobe
    C. Frontal lobe
    D. Parietal lobe
5. **Which CNS glial cell supports the blood brain barrier and regulates the ions around neurons?**
    A. Ependymal cell
    B. Astrocyte
    C. Microglia
    D. Oligodendrocyte
6. **After a stroke, a patient speaks fluently but the words make little sense, and he cannot understand what others say. The damage is most likely in:**
    A. Wernicke's area in the temporal lobe
    B. Broca's area in the frontal lobe
    C. Primary visual cortex
    D. The precentral gyrus
7. **A sensory neuron is stimulated more strongly. Since action potential size cannot change, how is the greater intensity encoded?**
    A. Larger amplitude of each action potential
    B. Higher frequency of action potentials
    C. Longer duration of each action potential
    D. Slower conduction along the axon

8. **The enteric nervous system is located in the:**
    A. Cerebral cortex
    B. Wall of the digestive tract
    C. Dorsal root ganglia
    D. Walls of the heart chambers
9. **A drug blocks the reuptake of serotonin into presynaptic terminals. What is the immediate effect at the synapse?**
    A. Serotonin synthesis stops
    B. Serotonin receptors are blocked
    C. Serotonin is broken down faster
    D. Serotonin stays in the cleft longer
10. **With each ATP used, the Na+/K+ pump moves:**
    A. 2 Na+ out and 3 K+ in
    B. 3 Na+ in and 2 K+ out
    C. Equal amounts of Na+ and K+ out
    D. 3 Na+ out and 2 K+ in
11. **The superior colliculi of the midbrain are mainly involved in:**
    A. Visual reflexes such as tracking objects
    B. Auditory reflexes such as turning toward sound
    C. Setting the rhythm of breathing
    D. Relaying taste to the cortex
12. **The ventral (anterior) horns of spinal cord gray matter contain:**
    A. Cell bodies of sensory neurons
    B. Ascending sensory tracts
    C. Cell bodies of somatic motor neurons
    D. Sympathetic preganglionic neurons

Answer key

1. **C.** Initiate and regulate movement. The basal nuclei work with the cortex and substantia nigra to start wanted movements and suppress unwanted ones. Relaying sensory input to the cortex is the thalamus.
2. **B.** A thick myelinated motor axon. Conduction velocity rises with both larger diameter and myelination, so a thick myelinated axon is fastest. A thick axon that lost its myelin must use slow continuous conduction and may fail entirely.
3. **A.** K+ leaking out through leak channels. The resting membrane has many more K+ leak channels than Na+ leak channels, so positive K+ ions drift out and leave the inside negative. Na+ leaking in actually makes the inside slightly less negative, so it cannot be the main cause.
4. **C.** Frontal lobe. Broca's area lies in the inferior frontal lobe, usually on the left, near the motor cortex for the face. Wernicke's area, for understanding language, is in the temporal lobe.
5. **B.** Astrocyte. Astrocytes wrap capillaries, help induce the blood brain barrier and buffer extracellular K+ and neurotransmitters. Ependymal cells line the ventricles and help make CSF instead.

6. **A.** Wernicke's area in the temporal lobe. Wernicke's area handles language comprehension, so damage produces fluent but meaningless speech and poor understanding. Broca's area damage instead gives slow, effortful speech with fairly good comprehension.
7. **B.** Higher frequency of action potentials. Stimulus strength is coded by how often action potentials fire (and how many neurons are recruited). Amplitude is fixed by the all or none principle, so a larger spike is not possible.
8. **B.** Wall of the digestive tract. The enteric nervous system is a large network of neurons in the wall of the gut that controls motility and secretion. The heart is influenced by autonomic nerves but has no enteric system.
9. **D.** Serotonin stays in the cleft longer. Reuptake is the main way serotonin is cleared, so blocking it prolongs its action on receptors, which is how SSRI antidepressants work. The drug does not block the receptors themselves.
10. **D.** 3 Na+ out and 2 K+ in. The pump exports 3 Na+ and imports 2 K+ per ATP, maintaining the concentration gradients and adding a small negative charge inside. Reversing the numbers (2 out, 3 in) is a common mix up.
11. **A.** Visual reflexes such as tracking objects. The superior colliculi coordinate visual reflexes and eye and head movements. The inferior colliculi, just below them, handle auditory reflexes.
12. **C.** Cell bodies of somatic motor neurons. Ventral horns hold the motor neurons that send axons out the ventral root to skeletal muscle. Sympathetic preganglionic cell bodies are in the lateral horns, and sensory cell bodies are in dorsal root ganglia.

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